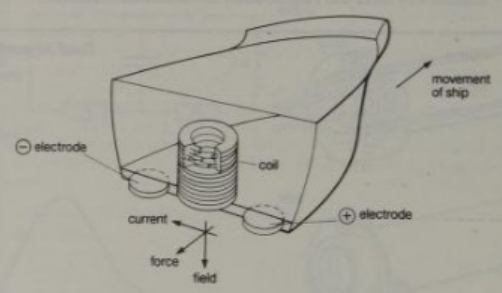
37 A ship without a propeller

Powered ships normally use a propeller system to enable them to move. The forward force on the ship is provided by giving backward momentum to the water. Various forms of engine, such as steam turbines or gas turbines may be used, but they tend to be only about 20% efficient. Whatever type of engine is used, the propeller system is inefficient, because it is converting circular motion into linear motion; the large amount of turbulence in the water near a propeller means energy has been wasted.

A novel means of propulsion may be possible in the near future thanks to the advances in superconductor technology. Superconductors will be used to make powerful electromagnets, producing a magnetic field in the sea water below the ship. At the same time an electric current will be passed through the sea water from electrodes on the bottom of the ship's hull. The current is acted on by the field, producing a force on the water in one direction, and a corresponding force on the ship in the other, as shown in the diagram.



Of course the ship must have some sort of electrical generator, but the shaft and propeller are no longer needed. What is more, placing electrodes in various positions on the bottom of the hull and switching the current to selected pairs of them can produce a force in any direction, so the ship can be steered. The rudder is therefore unnecessary too.

As with most advances, there may be a price to pay. The passage of current will electolyse the sea water, producing potentially harmful chemical concentrations. Also the large currents created in the sea water will warm the water up though given the vast volume of the sea, the rise in temperature is unlikely to be serious.

(a) If a propeller can make 100 kg of water each second move at $20\,\mathrm{ms^{-1}}$, how much force will it produce? (2)

(b) Why will the force calculated in (a) reduce as the ship gains speed? (2)

(c) What happens to the resistance force (drag) of the ship through the water as it speeds up? Under what conditions does a ship reach its maximum speed? (4)

(d) In the electromagnetic form of propulsion, what are the charge carriers in the sea water? (2)

(e) Sketch the sort of path you would expect the current to follow through the sea water. (3)

(f) Write down an expression for the force on a conductor of length l carrying a current I through a uniform magnetic field strength B. Why is this expression not strictly applicable to the situation shown in the diagram? (4)

(g) For a large ship a large force will be needed, say 2×10^6 N. If the distance between the electrodes is 50 m and the strength of the field is 10 T, calculate the current needed to produce this force. (Assume the expression in (f) is applicable.) (3)

(h) Where would you place the positive and negative electrodes to produce a force which would steer the ship to the left? (3)

(i) Using alternating current between the electrodes would almost eliminate the electrolysis of the sea water, but would also produce no net driving force! Explain why there would be no force, and suggest another change to the system which would allow a.c. to be used effectively. (4)

(j) Electromagnetic machines normally have an iron core to increase the field. Iron saturates in a field of about 2 tesla. Explain why superconducting electromagnets do not have a core. (3)

Total 30 marks

69

CIE 9702 Physics (Paper 4 style) — Electromagnetic Propulsion of a Ship

A research team is testing a prototype ship that has **no propeller**. Instead, two flat metal electrodes are fixed beneath the hull and a strong magnetic field is produced perpendicular to the seawater between them.

When a current passes through the seawater, the water experiences a magnetic force. By Newton's third law, the ship experiences an equal and opposite force and moves forward. This method avoids moving propeller blades but requires large electric currents.

In one trial, the following design data are used:

- distance between electrodes $d = 0.50\,\mathrm{m}$
- magnetic flux density between electrodes $B = 12\,\mathrm{T}$
- electrode immersion depth gives effective conductor length in field $L = 1.8\,\mathrm{m}$
- seawater density $\rho = 1025\,\mathrm{kgm^{-3}}$

- ship mass (including payload) $m = 1.9 \times 10^5\,\text{kg}$

At different operating currents, the measured thrust force F is:

Current I / A	Thrust force F / N
4.0×10^4	8.6×10^5
6.0×10^4	1.29×10^6
8.0×10^4	1.73×10^6
1.0×10^5	2.15×10^6

The water drag force can be modeled by

$$F_{\text{drag}} = kv^2$$

where $k = 4.9 \times 10^4\,\text{kgm}^{-1}$ and v is ship speed.

Questions

(a) State the physical principle that explains why the ship moves forward when the water is forced backward by the electromagnetic interaction. [2]

(b) For a straight conductor in a magnetic field, the force magnitude is given by

$$F = BIL.$$

Use this expression to calculate the expected thrust when $I = 8.0 \times 10^4\,\text{A}$. Comment on whether your value is consistent with the measured data table. [4]

(c) Use the data table to determine the gradient of a graph of F against I. Hence estimate an experimental value for BL. Compare this with the design values and suggest one reason for any difference. [5]

(d) At a speed of $6.0\,\text{ms}^{-1}$, calculate:

(i) the drag force,

(ii) the resultant forward force when $I = 1.0 \times 10^5\,\text{A}$,

(iii) the acceleration of the ship.

[5]

(e) During operation at constant current, speed increases and then approaches a steady value. Explain why terminal speed is reached. Your answer should refer to how drag and thrust vary with speed. [3]

(f) The potential difference across the seawater gap is approximately

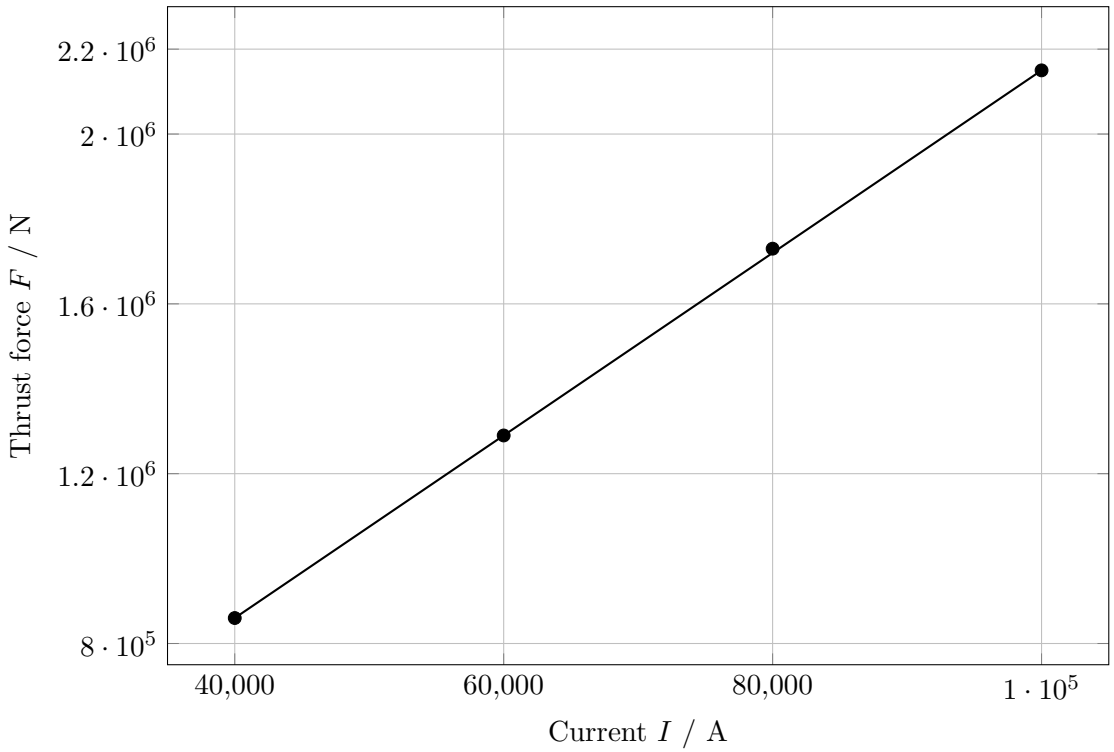
$$V = Ed,$$

where E is the electric field between electrodes. If $E = 220\,\mathrm{Vm}^{-1}$, calculate the electrical power transferred to the seawater when $I = 1.0 \times 10^5\,\mathrm{A}$. [3]

(g) Electrolysis in seawater can produce chlorine-containing compounds and hydrogen gas. Describe two environmental or engineering concerns caused by electrolysis in this propulsion system. [2]

(h) Open-ended evaluation question. The company wants to replace conventional propellers on short-range ferries with electromagnetic propulsion. Using ideas from forces, efficiency, materials, environmental impact and controllability, discuss whether this is practical. Include at least one justified improvement to the prototype design. [6]

Graph of provided data



Suggested answers (outline)

(a) Newton's third law: force on water and equal opposite force on ship.

(b)

$$F = BIL = (12)(8.0 \times 10^4)(1.8) = 1.73 \times 10^6\,\mathrm{N}.$$

This agrees closely with the measured value $1.73 \times 10^6\,\mathrm{N}$.

(c) Gradient from table:

$$\frac{\Delta F}{\Delta I} = \frac{2.15 \times 10^6 - 8.6 \times 10^5}{1.0 \times 10^5 - 4.0 \times 10^4} = 21.5\,\mathrm{N\ A^{-1}}.$$

Since $F = (BL)I$, gradient $= BL$, so experimental $BL \approx 21.5$. Design $BL = (12)(1.8) = 21.6$, very close. Possible differences: non-uniform field, electrode polarization, measurement uncertainty, turbulence.

(d)

$$F_{\mathrm{drag}} = kv^2 = (4.9 \times 10^4)(6.0)^2 = 1.76 \times 10^6\,\mathrm{N}.$$

At $I = 1.0 \times 10^5\,\mathrm{A}$, thrust $\approx 2.15 \times 10^6\,\mathrm{N}$.

$$F_{\mathrm{resultant}} = 2.15 \times 10^6 - 1.76 \times 10^6 = 3.9 \times 10^5\,\mathrm{N}.$$

$$a = \frac{F}{m} = \frac{3.9 \times 10^5}{1.9 \times 10^5} = 2.1\,\mathrm{m\ s^{-2}}.$$

(e) As v increases, $F_{\text{drag}} = kv^2$ rises rapidly while thrust is approximately fixed for fixed current. Terminal speed occurs when drag equals thrust, so resultant force and acceleration become zero.

(f)

$$V = Ed = (220)(0.50) = 110\,\text{V}, \qquad P = VI = (110)(1.0 \times 10^5) = 1.1 \times 10^7\,\text{W}.$$

So power is 11 MW.

(g) Any two: toxic by-products in seawater, corrosion of electrodes/hull, gas bubble formation reducing efficiency, safety risks from hydrogen evolution.

(h) Indicative high-level points: large controllable thrust and no moving propeller parts are advantages; however very high current and strong superconducting magnets increase cost and energy demand. Electrolysis and corrosion are major engineering/environmental issues. Practical use may be limited to specialized vessels unless efficiency improves. Reasoned improvements may include pulse-current control, optimized electrode coatings, closed-loop power electronics, or hybrid propeller-electromagnetic operation.

Variant 2 (Higher Difficulty)

Passage

An upgraded electromagnetic ferry is tested on a coastal route. The design team uses segmented electrodes and variable-current control to improve thrust at low speed while limiting thermal damage. The ship still uses seawater as the conducting fluid, and the magnetic field is generated by superconducting coils beneath the hull.

The ferry has mass $m = 2.40 \times 10^5\,\text{kg}$. The magnetic field in the propulsion channel is $B = 14.0\,\text{T}$ and the effective conductor length is $L = 2.10\,\text{m}$. Because of field non-uniformity and bubble formation, only a fraction η_m of the ideal magnetic force contributes to useful thrust:

$$F_{\text{thrust}} = \eta_m BIL.$$

During this test, $\eta_m = 0.88$.

The opposing water resistance is modeled by

$$F_{\text{drag}} = av + bv^2$$

where $a = 3.2 \times 10^4\,\text{Nsm}^{-1}$ and $b = 3.8 \times 10^4\,\text{kgm}^{-1}$.

At current $I = 9.0 \times 10^4\,\text{A}$, measured terminal speed is $6.8\,\text{ms}^{-1}$.

Additional operating data for constant-field operation are shown below.

Current I / A	Voltage V / V	Coolant power / MW	Measured thrust / N
6.0×10^4	95	1.8	1.55×10^6
8.0×10^4	112	2.4	2.07×10^6
1.0×10^5	131	3.1	2.60×10^6
1.2×10^5	153	4.0	3.11×10^6

Questions (Variant 2)

(a) Show that the ideal magnetic thrust at $I = 1.0 \times 10^5\,\text{A}$ is approximately $2.94 \times 10^6\,\text{N}$. Hence calculate the useful thrust after accounting for η_m. [3]

(b) For $I = 9.0 \times 10^4\,\text{A}$ and speed $v = 6.8\,\text{ms}^{-1}$, calculate:

(i) F_{drag} using the linear-plus-quadratic model,

(ii) the corresponding useful magnetic thrust,

(iii) whether the ferry should be accelerating, decelerating, or at terminal speed.

Comment on consistency with the measured terminal-speed statement. [6]

(c) Using the operating data table, calculate the electrical input power $P_{\text{in}} = VI$ for each current and determine the propulsion efficiency

$$\eta_p = \frac{F_{\text{thrust}} v}{P_{\text{in}} + P_{\text{cool}}}$$

at $v = 6.8\,\text{ms}^{-1}$ for the $I = 1.0 \times 10^5\,\text{A}$ case. Take P_{cool} from the table. [5]

(d) Plot or use the provided graph of measured thrust against current to determine whether the relationship is linear over this range. State one physical reason why slight non-linearity may occur at high current. [4]

(e) The electrodes are separated by $d = 0.60\,\text{m}$. Assuming a uniform field between electrodes, calculate the seawater conductivity σ for the $I = 8.0 \times 10^4\,\text{A}$ operating point if the active cross-sectional area of current flow is $A = 2.5\,\text{m}^2$. Use

$$J = \sigma E, \qquad J = \frac{I}{A}, \qquad E = \frac{V}{d}.$$

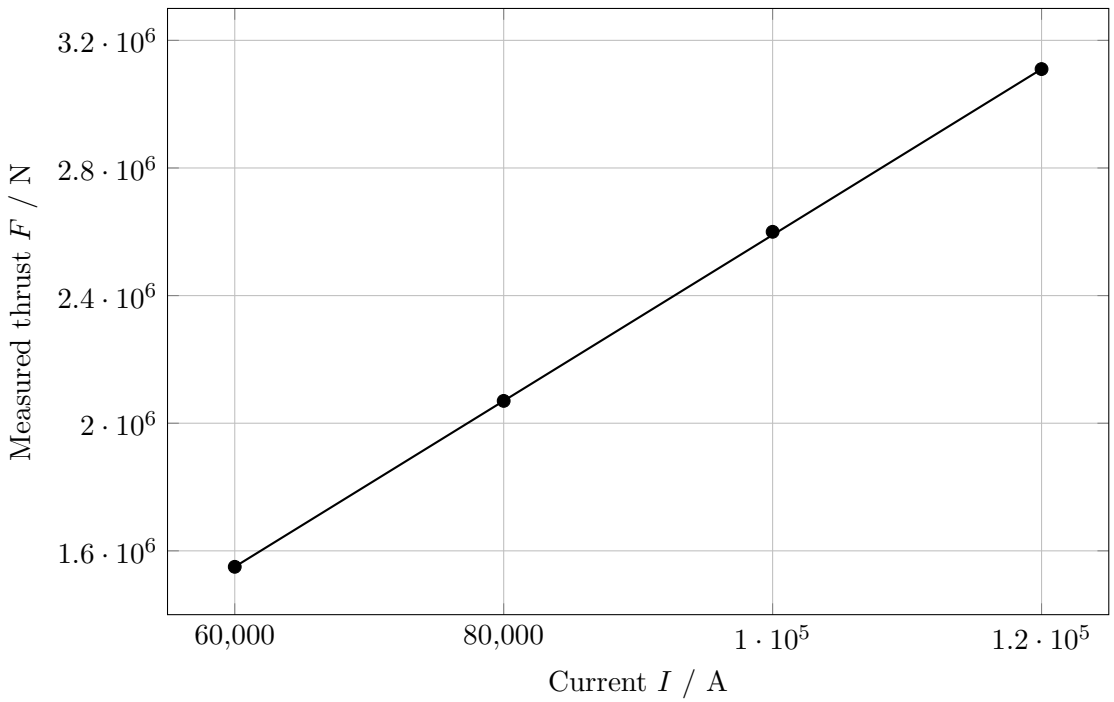
[5]

(f) Open-ended synthesis question. The design team can either:

- increase magnetic field strength by 20% (higher cryogenic load), or
- increase electrode area by 20% (larger hull integration and mass penalty).

Evaluate which option is more effective for improving real-world performance on short ferry routes. Your answer should include quantitative support, engineering trade-offs, and environmental implications. [7]

Variant 2 graph (measured thrust vs current)



Suggested answers (Variant 2 outline)

(a)

$$F_{\text{ideal}} = BIL = (14.0)(1.0 \times 10^5)(2.10) = 2.94 \times 10^6\,\text{N}.$$
$$F_{\text{useful}} = \eta_m F_{\text{ideal}} = (0.88)(2.94 \times 10^6) = 2.59 \times 10^6\,\text{N}.$$

(b)

$$F_{\text{drag}} = av + bv^2 = (3.2 \times 10^4)(6.8) + (3.8 \times 10^4)(6.8)^2 \approx 1.97 \times 10^6\,\text{N}.$$
$$F_{\text{thrust}} = (0.88)(14.0)(9.0 \times 10^4)(2.10) \approx 2.33 \times 10^6\,\text{N}.$$

Resultant force $\approx 3.6 \times 10^5$ N forward, so model predicts slight acceleration. If measured speed is terminal, possible causes include wave effects, effective η_m drop at speed, or parameter uncertainty.

(c) For $I = 1.0 \times 10^5$ A:

$$P_{\text{in}} = VI = (131)(1.0 \times 10^5) = 1.31 \times 10^7\,\text{W}.$$

$$P_{\text{cool}} = 3.1 \times 10^6\,\text{W}, \quad P_{\text{mech}} = Fv = (2.60 \times 10^6)(6.8) = 1.77 \times 10^7\,\text{W}.$$
$$\eta_p = \frac{1.77 \times 10^7}{1.31 \times 10^7 + 3.1 \times 10^6} \approx 1.09.$$

This unphysical value suggests inconsistent assumptions (e.g. thrust measured at different speed/conditions) and requires critical discussion.

(d) Data are close to linear with small deviation at highest current. Possible reasons: magnet saturation in return path, increased bubble fraction, heating effects changing conductivity.

(e) At $I = 8.0 \times 10^4\,\mathrm{A}$ and $V = 112\,\mathrm{V}$:

$$J = \frac{I}{A} = \frac{8.0 \times 10^4}{2.5} = 3.2 \times 10^4\ \mathrm{A\ m^{-2}}, \quad E = \frac{V}{d} = \frac{112}{0.60} = 1.87 \times 10^2\ \mathrm{V\ m^{-1}}.$$

$$\sigma = \frac{J}{E} \approx 1.7 \times 10^2\ \mathrm{S\ m^{-1}}.$$

(f) Indicative evaluation points: increasing B directly increases thrust but causes major cryogenic and magnet-cost penalties; larger electrode area reduces current density and electrolysis severity but may increase drag and hull mass. Best answers compare lifecycle efficiency, reliability, maintenance and emissions impacts, then justify a context-dependent recommendation.